Active Maintenance Report: coppock_2019b

Coppock (2019, Journal of Experimental Political Science): Avoiding Post-Treatment Bias in Audit Experiments

2026-08-01

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Archived at Zenodo: doi:10.5281/zenodo.24168312. This DOI always resolves to the latest release of the maintained version.

Drafted by Claude Opus 5 under the supervision of Alex Coppock.

This repository holds the actively maintained replication code for Coppock (2019), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1017/XPS.2018.9
Replication archive	10.7910/DVN/6NVI9C
Underlying data (White, Nathan and Faller 2015)	10.7910/DVN/28158

The data are not redistributed here. The deposit is 1.3 MB across four files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or set of in-text quantities, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it: `published_claims.csv` is the extraction of every number the article prints, `run_archive.R` runs the deposited script exactly as shipped and records in `archive_run_status.csv` where it stops, `extract_archive_values.R` runs it again with its defects repaired, and `build_ground_truth.R` assembles the comparison, writes `float_coverage.csv`, and gates both. `original/` is created by the download script and is deliberately absent from the repository. `errata.qmd` renders the four corrections the article needs. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_2019b.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its four files, produces the published table and every in-text quantity into `maintained/output/`, runs the deposited script in a scratch directory, rebuilds the ground truth with its coverage gates, and prints one line per numeric claim the article

makes. Required packages: tidyverse, estimatr, rsample, broom, xtable, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The run takes well under a minute, most of it in the 4,000 bootstrap replicates of the bounds estimator. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.**

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

No. The archive's single script stops partway through on a current R installation, and it stops twice, at two independent points, neither of which produces a helpful error message.

The first failure is the bootstrap. `broom::bootstrap()` still exists, but it builds a grouped data frame in the format dplyr used before version 0.8.0, and the `dplyr::do()` on the next line rejects it with `Corrupt grouped_df using old (< 0.8.0) format`. The second failure is downstream of that and would be reached by anyone who fixed the first: the script reads `fit_1$coefficients`, `fit_1$ci_lower` and `fit_1$ci_upper` out of a tidied `lm_robust` object, and current `estimatr` names those columns `estimate`, `conf.low` and `conf.high`. That one fails with `non-numeric argument to mathematical function`, which says nothing about the renamed column. A third defect stops the script on its first line for anyone not working on macOS or Windows: the `load()` call asks for `voterIDexp_data_sept2014.Rdata` and the deposited file is `voterIDexp_data_sept2014.RData`.

Both are mechanical and resolvable without the author. Ten lines in four places make it run to completion, and 27 of the 31 published values it can be checked against come back. The 4 that do not are described below, and none of the four is the archive's fault: two belong to R and two to the article's own prose.

1.2 Does the maintained rewrite reproduce the paper?

Yes, on every quantity the paper's own analysis produces. Every estimate, confidence interval and standard error in Table 2 and in the article's prose reproduces to the precision the article prints, and so do the three response rates on the first page, which the deposited script never computes at all. Both of the article's descriptive claims about statistical significance hold.

4 of the 31 checkable rows carry `match_rewrite = 0`, and none of the four is a reproduction failure.

Two are the endpoints of the bootstrap confidence interval on the bounds. R 3.6.0 changed how `sample()` converts uniform draws into integers, so the archive's `set.seed(343)` selects different resamples than it did in 2018. Restoring the old sampler with `RNGkind(sample.kind = "Rounding")` returns the published interval exactly. The drift is the sampler, not the bootstrap package: the two are shown below to draw identically.

The other two are inconsistencies inside the article, and each is an entry in the errata note this repository carries. Table 2 reports the Always-Responder estimate as −0.056 and the text on the same page reports it as −5.5 points. The estimate is −5.563 points, so the table rounds and the text truncates. The same truncation appears one sentence later, where the standard error on the redefined outcome is stated as 1.7 points against an actual 1.756.

2 Paper overview

Citation: Coppock, A. (2019). "Avoiding post-treatment bias in audit experiments." *Journal of Experimental Political Science*, 6(1), 1-4. DOI: 10.1017/XPS.2018.9

Summary: Audit experiments randomize a requester characteristic, such as an apparently Latino versus an apparently non-Latino White name, and measure whether a request draws a response. Researchers routinely want to go further and ask whether the responses that arrive are better or worse, but response is itself a consequence of treatment, so conditioning on it de-randomizes the experiment. The note takes White, Nathan and Faller's (2015) audit of local election officials as a worked example and lays out the three options an analyst actually has: bound the effect among subjects who would respond either way, find a subgroup that can be assumed to respond either way, or redefine the outcome so that a non-response counts as zero. Applied to the same data, the naive conditional estimate is −3.5 points and not significant, while the redefined-outcome estimate is −6.1 points and the matched-pair Always-Responder estimate is −5.6 points, both significant. Conditioning on response does not merely bias the estimate in principle; here it attenuates it enough to change the conclusion.

3 Original archive reproducibility

The deposit is four files: the analysis script, the White et al. data it reanalyzes, a README, and an HTML log of a 2018 run. There is one script and it produces one table.

Table 1: Original archive reproducibility, checked against R 4.6.0 on 1 August 2026.

File	Status on current R	Resolution
aec_replication_JEPS.R	Fails at two points on current R	Replace the do() over broom::bootstrap(); rename three tidy() columns
voterID-exp_data_sept2014.RData	Loads clean	Correct the case of the filename in the load() call

aec_replication_JEPS.html	Log of a 2018 run; carries Table 2 but no standard errors	None needed
README.txt	Describes the files and the constructed variables	None needed

Every package the script names still installs from CRAN: `dplyr`, `estimatr`, `xtable` and `broom`. Nothing here was retired. All three defects are the same kind of thing, a name or a data structure that moved underneath code that was correct when it was written.

3.1 The bootstrap

The script draws its bootstrap replicates like this:

```
set.seed(343)
boot_df <-
  bootstrap(wnf_subset, 1000) %>%
  do(bounds_wrapper(.)) %>%
  ungroup()
```

`broom::bootstrap()` was deprecated in broom 0.5.0 and is still present in broom 1.0.13, where it warns and then returns a data frame carrying the `indices`, `labels` and `vars` attributes that dplyr used before version 0.8.0. Modern dplyr wants a `groups` tibble instead, so `do()` refuses the object it is handed. The function is not missing and the error does not mention it; the visible failure is two calls downstream of the cause.

3.2 The renamed tidy columns

Once the bootstrap is repaired the script fails again, at the table assembly:

```
estimates <- c(round(fit_1$coefficients[2], 3), ...)
cis <- c(paste0("[", round(fit_1$ci_lower[2], 3), ", ", ...))
```

`fit_1` is `tidy(lm_robust(...))`. Current `estimatr` returns that tidy frame with columns `estimate`, `conf.low` and `conf.high`, following broom's conventions; the names the script uses were the ones estimatr returned in 2018. Indexing a data frame by a column that does not exist returns `NULL`, and `round(NULL, 3)` is the error the user actually sees.

3.3 The filename

`load("voterIDexp_data_sept2014.Rdata")` asks for a file the deposit spells `voterIDexp_data_sept2014.RDat`
It resolves on macOS and Windows, whose filesystems are case-insensitive by default, and fails on the first line for anyone running on Linux. This defect is invisible to the author and to any reviewer working on the same platform, which is exactly what makes it worth recording.

4 Errata

The article needs four corrections, three of them in the reanalysis paragraph on page 3 and the fourth in Table 2, and none of them changes a conclusion. `errata.qmd` sets them out and renders to `coppock_2019b_errata.pdf` at the root of this repository, computing every corrected value from `maintained/output/` at the moment it is rendered, so the note cannot go stale in the way the sentences did.

Two of them are the same defect. The article gives the Always-Responder estimate as −5.5 points where its own Table 2 gives −0.056, and the standard error on the redefined outcome as 1.7 points where the standard error is 1.756. In each case the published figure is the correct value truncated rather than rounded at the last digit it prints. Table 2 is right in every cell it prints, which is why the errata quotes its cells rather than reprinting it.

The third is a dropped word rather than a wrong number. The same paragraph gives the naive estimate as −3.5 percentage, with no unit, where every other figure around it is in points. It came out of reading the article word by word against its own PDF to prepare a remastered edition, which is a finer sieve than the number-by-number comparison below.

That is a separate matter from the repairs described above, which are to the deposited *code*. No analytical decision in the deposit was changed, and the rewrite corrects no coding error, because there is none to correct.

5 Number-by-number comparison

Table 2: Ground truth: published value against the value the repaired archive script produces on current R. Published values are read from the article; a blank Archive column means the deposited script never computes the quantity.

Location	Claim	Paper	Archive	Match
Introduction, p. 1	Share of putatively non-Latino White names that received a response	70.5	70.50720000	1
Introduction, p. 1	Share of putatively Latino names that received a response	64.8	64.83100000	1
Introduction, p. 1	Difference in response rates, percentage points	-5.7	-5.67616000	1

Table 2: Ground truth: published value against the value the repaired archive script produces on current R. Published values are read from the article; a blank Archive column means the deposited script never computes the quantity. *(continued)*

Location	Claim	Paper	Archive	Match
Introduction, p. 1	Whether the difference in response rates is statistically significant at the 0.05 level	1	NA	
Table 1, p. 2	Table 1, Always-Responder, R_i(0)	1	1.00000000	1
Table 1, p. 2	Table 1, Always-Responder, R_i(1)	1	1.00000000	1
Find Always-Responders, p. 2	Response rate required in both arms under the Always-Responders assumption	100	100.00000000	1
Table 2, p. 3	Table 2, naive difference-in-means, estimate	-0.035	-0.03542390	1
Table 2, p. 3	Table 2, naive difference-in-means, lower confidence limit	-0.075	-0.07544890	1
Table 2, p. 3	Table 2, naive difference-in-means, upper confidence limit	0.005	0.00460118	1
Table 2, p. 3	Table 2, redefined outcome, estimate	-0.061	-0.06099950	1
Table 2, p. 3	Table 2, redefined outcome, lower confidence limit	-0.095	-0.09543510	1
Table 2, p. 3	Table 2, redefined outcome, upper confidence limit	-0.027	-0.02656380	1
Table 2, p. 3	Table 2, matched-pair Always-Responders, estimate	-0.056	-0.05563280	1
Table 2, p. 3	Table 2, matched-pair Always-Responders, lower confidence limit	-0.104	-0.10431800	1
Table 2, p. 3	Table 2, matched-pair Always-Responders, upper confidence limit	-0.007	-0.00694736	1
Table 2, p. 3	Table 2, bounds, lower bound	-0.658	-0.65780100	1
Table 2, p. 3	Table 2, bounds, upper bound	0.645	0.64539000	1
Table 2, p. 3	Table 2, bounds, lower bootstrap confidence limit	-0.73	-0.73863800	0
Table 2, p. 3	Table 2, bounds, upper bootstrap confidence limit	0.726	0.72535200	0
Table 2, p. 3	Confidence level of the Table 2 intervals	95	95.00000000	1
Redefine the outcome, p. 3	Value the redefined outcome takes when no e-mail is sent	0	0.00000000	1
Reanalysis, p. 3	Zhang and Rubin lower bound, percentage points	-66	-65.78010000	1
Reanalysis, p. 3	Zhang and Rubin upper bound, percentage points	65	64.53900000	1
Reanalysis, p. 3	Matched pairs in which both members responded	719	719.00000000	1
Reanalysis, p. 3	Always-Responder ATE, percentage points	-5.5	-5.56328000	0
Reanalysis, p. 3	Always-Responder standard error, percentage points	2.5	2.48190000	1
Reanalysis, p. 3	Redefined-outcome ATE, percentage points	-6	-6.09995000	1
Reanalysis, p. 3	Redefined-outcome standard error, percentage points	1.7	1.75629000	0
Reanalysis, p. 3	Whether the redefined-outcome interval excludes zero while the naive interval contains it	1	NA	
Reanalysis, p. 3	Naive ATE, percentage points	-3.5	-3.54239000	1
Reanalysis, p. 3	Naive standard error, percentage points	2.0	2.04099000	1
Reanalysis, p. 3	Estimates displayed in Table 2	4	4.00000000	1

Table 2: Ground truth: published value against the value the repaired archive script produces on current R. Published values are read from the article; a blank Archive column means the deposited script never computes the quantity. *(continued)*

Location	Claim	Paper	Archive	Match

Of the 33 recorded claims, 31 can be compared against something the repaired archive produces. 27 match and 4 do not. The other 2 are the article's two claims about statistical significance, which have no number to compare and carry a truth value instead; both hold.

`value_script` is not typed here. `ground_truth/extract_archive_values.R` copies the deposit into a scratch directory, applies the four repairs described above to its own source text, runs it there, and writes what it produces to `ground_truth/archive_values.csv`, which is what this column reads. Three of the values in that file are counts and rates of the two frames the deposited cleaning code builds rather than results the deposited script prints: the article's three opening response rates are among them, and recording them as the deposit's own is the claim that the deposit's data support the figures the article attributes to White, Nathan and Faller (2015).

6 The extraction and the two instruments

`ground_truth/published_claims.csv` is the extraction: every numeric token in the article's four published pages, read out with its surrounding context and then classified by hand, followed by a second sweep for numbers written as words, which no token scan sees. Citation years, DOI numerals, running heads and the publisher's own front matter are the only tokens excluded. The article has no separate appendix.

Table 3: The extraction, by claim type. A pipeline or descriptive claim must have a ground-truth row and a block in the claims file; a definitional, structural or transcribed claim gets a block only where the pipeline can reach the quantity, and is otherwise verified at the point of use.

Class	Claims	Requiring a block
definitional	15	5
descriptive	2	2
pipeline	25	25
structural	6	1
transcribed	4	0

Coverage of the two published tables is stated as a fraction rather than as an existence check, because one row is not coverage of a float. All 14 of Table 2's published numbers are covered, counting its confidence level. 2 of Table 1's 8 are: Table 1 is a stipulated typology of principal

strata, and only the Always-Responder row names units any analysis identifies. The other three strata are verified nowhere, which is a property of the paper rather than a gap in this repository.

`maintained/in_text_claims.R` is the second instrument. It carries one block per claim, each with the article's own sentence quoted verbatim above code that recomputes the number from `maintained/output/` and prints it at the precision the article prints. It never reads the ground truth, never refits a model, and where `build_ground_truth.R` reaches a quantity through one output file it reaches the same quantity through another: the percentage-point figures come from the Table 2 estimates rather than from `text_in_text_estimates.csv`, the pair count and the response rates from the analysis frames, and the two significance claims from the confidence intervals where the ground truth uses the p-values. Agreement between two derivations is evidence the arithmetic is right; it is not evidence that the quantity is the one the sentence names, which is what writing the block out of the sentence is for.

`build_ground_truth.R` then gates the whole arrangement. It sources the claims file as a program into its own environment and counts the claim lines it prints, so a block that errors, or that ends in a bare expression and prints nothing under `source()`, fails rather than passing as coverage. It asserts that the 33 printed claim ids are exactly the 33 the extraction declares, that every pipeline and descriptive claim has a ground-truth row, that no ground-truth row names a claim the article does not make, and that the two instruments agree value by value at the article's own precision. It also asserts that every row carrying an adverse verdict has a `defect_locus` and that no clean row has one, reading all three of `match`, `match_rewrite` and `holds`, since a gate stated on the rewrite alone cannot see a failing descriptive claim or an archive failure the rewrite passes.

Table 4: Every row whose verdict is adverse, and where the fault lies.

Location	Claim	Paper	Rewrite	Locus
Table 2, p. 3	Table 2, bounds, lower bootstrap confidence limit	-0.73	-0.739	environment
Table 2, p. 3	Table 2, bounds, upper bootstrap confidence limit	0.726	0.725	environment
Reanalysis, p. 3	Always-Responder ATE, percentage points	-5.5	-5.563	paper_internal
Reanalysis, p. 3	Redefined-outcome standard error, percentage points	1.7	1.756	paper_internal

2 are the environment, and 2 are the article disagreeing with itself. Neither is a defect in the deposited code and neither is a defect in the rewrite.

7 The bootstrap interval and the R 3.6 sampler

The one published quantity the archive no longer returns is the bootstrap confidence interval on the bounds, and the reason is not in the archive.

R 3.6.0, released in April 2019, replaced the "Rounding" method that `sample()` used to convert uniform variates into integers with a "Rejection" method. Every resample the bootstrap draws therefore differs from the one the same seed drew in 2018. `text_bootstrap_sampler.R` computes the interval three ways to isolate the cause.

Table 5: The bounds bootstrap interval, computed three ways, against the published values. Only the pre-3.6 sampler returns the article's interval.

Method	Lower	Upper
Maintained rewrite: rsample::bootstraps(), R 3.6+ sampler	-0.7386	0.7254
Archive resampling scheme, R 3.6+ sampler	-0.7386	0.7254
Archive resampling scheme, pre-3.6 sampler	-0.7296	0.7258
Published (Table 2)	-0.73	0.726

Two things follow. The first is that `rsample::bootstraps()` and the archive's own `sample(n, replace = TRUE)` scheme give the identical interval at the identical seed, to the last digit. The two rows above that agree exactly are the same 1,000 resamples reached by two routes, so no difference between the bootstrap implementations can explain the drift.

The second is that restoring the old sampler restores the published interval to the digit the article prints, $[-0.73, 0.726]$. The deposit is internally reproducible. What changed was R.

The maintained rewrite keeps the modern sampler, because reproducing a number by pinning a retired random number generator is a way of preserving an artifact rather than a result. The difference is in any case well inside Monte Carlo noise for a 1,000-replicate bootstrap of an interval more than 1.4 wide.

8 Maintained rewrite

The rewrite lives in `maintained/`: a shared `helpers.R`, one cleaning script, one table script, and three scripts for quantities the article states in prose. It is a translation, not a reanalysis: every estimator, specification and sample restriction is the one the paper used.

Table 6: Maintained rewrite scripts.

Script	Output	Description
helpers.R	(sourced by every script)	Packages and the Zhang and Rubin bounds estimator

clean_voterID.R	wnf_subset.rds, wnf_always_responders.rds	The two analysis frames: week-one non-bounced emails, and matched-pair Always-Responders
table_2_reanalysis.R	table_2_reanalysis.csv	The four estimators of Table 2, at full precision
text_in_text_estimates.R	text_in_text_estimates.csv	The bounds, the pair count and each estimator in percentage points
text_response_rates.R	text_response_rates.csv	The response rates the first page attributes to White et al. (2015)
text_bootstrap_sampler.R	text_bootstrap_sampler.csv	The bounds interval under three resampling routes
in_text_claims.R	(printed claim lines only)	One block per numeric claim the article makes, recomputed and printed

8.1 Architecture

`clean_voterID.R` builds both analysis frames and writes them to `output/`, so the sample restrictions are stated once and every downstream script reads the same rows. `table_2_reanalysis.R` writes its estimates at full precision rather than at the three decimals the article prints, and the two in-text scripts read that file rather than recomputing the models. Nothing in this repository rounds a number twice, and no published value is typed into a script.

The bounds estimator is the one place where the rewrite reimplements rather than replaces. It is the Zhang and Rubin (2003) trimming estimator, which the archive adapted from Aronow, Baron and Pinson's source code; it exists in no package, so `helpers.R` carries it, translated from base-R indexing and two accumulator loops into `arrange`, `filter` and `cumsum`. It is called from both the table script and the sampler script, which is why it lives in `helpers.R` and not in either.

`text_response_rates.R` is an addition rather than a translation. The article's opening paragraph reports that non-Latino White names drew a response 70.5 percent of the time and Latino names 64.8 percent, a difference of −5.7 points, attributing the figures to White et al. (2015). The deposited script never computes them, so nothing in the deposit shows they can be recovered. They can, and on exactly the analysis frame the rest of the paper uses.

Table 7: The first paragraph's response rates, recovered from the deposited data. The article states 70.5, 64.8 and -5.7.

Quantity	Rewrite	N
Response rate, Putatively non-Latino White name	70.5	1597
Response rate, Putatively Latino name	64.8	1598
Difference in response rates, percentage points	-5.7	3195

8.2 Deprecated patterns replaced

Table 8: Deprecated patterns and their replacements in the maintained rewrite.

Original pattern	Replacement
‘rm(list = ls())‘	(omitted)
‘load(”...Rdata”)‘	‘load(”...RData”)‘, the name the deposit carries
‘broom::bootstrap()‘ + ‘dplyr::do()‘	‘rsample::bootstraps()‘ + ‘purrr::map()‘
‘fit$coefficients‘, ‘fit$ci_lower‘, ‘fit$ci_upper‘	‘estimate‘, ‘conf.low‘, ‘conf.high‘
‘within()‘ blocks over base indexing	‘mutate()‘ paragraphs with ‘if_else()‘
‘left_join()‘ with no ‘by =‘	‘left_join(by = ”pair_id”)‘
‘data.frame()‘ inside the estimator	‘tibble()‘
two accumulator ‘for‘ loops	‘arrange()‘ + ‘cumsum()‘
‘xtable‘ + ‘print.xtable‘ (commented out)	‘write_csv()‘ to ‘output/‘
magrittr pipe	native pipe

`xtable` is left alone where a rewrite still needs it; here it is not needed, because the archive's only `xtable` call is commented out and the table it would have written is reproduced as a CSV.

9 Table 2 as reproduced

Table 9: Table 2 of the article, as reproduced by the maintained rewrite. The article prints the same four estimates; its bounds interval is [-0.73, 0.726], for the reason given above.

Estimand	Estimator	Estimate	95% CI
Undefined	Naive difference-in-means	-0.035	[-0.075, 0.005]
ATE on redefined outcome	Difference-in-means	-0.061	[-0.095, -0.027]
ATE among matched pairs in which both members are Always-Responders	Difference-in-means among matched pairs in which both members respond	-0.056	[-0.104, -0.007]
ATE among all Always-Responders	Bounds	[-0.658, 0.645]	[-0.739, 0.725]

Table 10: The quantities the article states in prose, at full precision. The article rounds the bounds to -66 and 65 points and the standard errors to one decimal.

Quantity	Rewrite
Lower bound, percentage points	-65.780
Upper bound, percentage points	64.539
Matched pairs in which both members responded	719.000
Always-Responder ATE, percentage points	-5.563
Always-Responder SE, percentage points	2.482
Redefined-outcome ATE, percentage points	-6.100
Redefined-outcome SE, percentage points	1.756
Naive ATE, percentage points	-3.542

Naive SE, percentage points	2.041

10 Maintained rewrite verification

Table 11: Maintained rewrite verification: published value against rewrite output. The verdict is the rewrite's match, except on the two descriptive claims, which have no number to compare and carry a truth value instead.

Location	Claim	Paper	Rewrite	Verdict	Note
Introduction, p. 1	Share of putatively non-Latino White names that received a response	70.5	70.50720000	1	The deposited script never computes this; value_script is the rate on the analysis frame the deposit itself builds
Introduction, p. 1	Share of putatively Latino names that received a response	64.8	64.83100000	1	The deposited script never computes this; value_script is the rate on the analysis frame the deposit itself builds
Introduction, p. 1	Difference in response rates, percentage points	-5.7	-5.67616000	1	The deposited script never computes this; value_script is the difference on the analysis frame the deposit itself builds
Introduction, p. 1	Whether the difference in response rates is statistically significant at the 0.05 level	1	1.00000000	1	The deposit fits no model of response, so it has no verdict here. The rewrite's two-sided p-value on the difference in response rates is the test the sentence claims
Table 1, p. 2	Table 1, Always-Responder, R_i(0)	1	1.00000000	1	Read back off the Always-Responder frame rather than asserted
Table 1, p. 2	Table 1, Always-Responder, R_i(1)	1	1.00000000	1	Read back off the Always-Responder frame rather than asserted
Find Always-Responders, p. 2	Response rate required in both arms under the Always-Responders assumption	100	100.00000000	1	The article's own stated check on the Always-Responders assumption
Table 2, p. 3	Table 2, naive difference-in-means, estimate	-0.035	-0.03542390	1	
Table 2, p. 3	Table 2, naive difference-in-means, lower confidence limit	-0.075	-0.07544890	1	
Table 2, p. 3	Table 2, naive difference-in-means, upper confidence limit	0.005	0.00460118	1	
Table 2, p. 3	Table 2, redefined outcome, estimate	-0.061	-0.06099950	1	
Table 2, p. 3	Table 2, redefined outcome, lower confidence limit	-0.095	-0.09543510	1	
Table 2, p. 3	Table 2, redefined outcome, upper confidence limit	-0.027	-0.02656380	1	
Table 2, p. 3	Table 2, matched-pair Always-Responders, estimate	-0.056	-0.05563280	1	
Table 2, p. 3	Table 2, matched-pair Always-Responders, lower confidence limit	-0.104	-0.10431800	1	
Table 2, p. 3	Table 2, matched-pair Always-Responders, upper confidence limit	-0.007	-0.00694736	1	
Table 2, p. 3	Table 2, bounds, lower bound	-0.658	-0.65780100	1	
Table 2, p. 3	Table 2, bounds, upper bound	0.645	0.64539000	1	
Table 2, p. 3	Table 2, bounds, lower bootstrap confidence limit	-0.73	-0.73863800	0	R 3.6.0 changed sample(); under RNGkind(sample.kind = 'Rounding') the same seed gives -0.730, which is the published value. text_bootstrap_sampler.csv carries all three intervals
Table 2, p. 3	Table 2, bounds, upper bootstrap confidence limit	0.726	0.72535200	0	R 3.6.0 changed sample(); under RNGkind(sample.kind = 'Rounding') the same seed gives 0.726, which is the published value. text_bootstrap_sampler.csv carries all three intervals

Table 11: Maintained rewrite verification: published value against rewrite output. The verdict is the rewrite's match, except on the two descriptive claims, which have no number to compare and carry a truth value instead. *(continued)*

Location	Claim	Paper	Rewrite	Verdict	Note
Table 2, p. 3	Confidence level of the Table 2 intervals	95	95.00000000	1	The column header is a claim about the intervals beside it; the level is recovered from the ratio of each half width to its standard error
Redefine the outcome, p. 3	Value the redefined outcome takes when no e-mail is sent	0	0.00000000	1	The redefined outcome's floor, read off the analysis frame
Reanalysis, p. 3	Zhang and Rubin lower bound, percentage points	-66	-65.78010000	1	The article rounds to whole points; -65.78 at full precision
Reanalysis, p. 3	Zhang and Rubin upper bound, percentage points	65	64.53900000	1	The article rounds to whole points; 64.54 at full precision
Reanalysis, p. 3	Matched pairs in which both members responded	719	719.00000000	1	Distinct pair identifiers among the Always-Responder pairs. One pair contributes an odd number of surviving rows, so halving the row count would give 719.5
Reanalysis, p. 3	Always-Responder ATE, percentage points	-5.5	-5.56328000	0	The text states -5.5 points and Table 2 states -0.056 for the same estimate. It is -5.563 points, so the table rounds and the text truncates. An inconsistency inside the article, not a reproduction failure
Reanalysis, p. 3	Always-Responder standard error, percentage points	2.5	2.48190000	1	2.482 at full precision
Reanalysis, p. 3	Redefined-outcome ATE, percentage points	-6	-6.09995000	1	The article states a 6 point decrease; -6.100 at full precision
Reanalysis, p. 3	Redefined-outcome standard error, percentage points	1.7	1.75629000	0	The text states SE = 1.7 points. It is 1.756 points, which rounds to 1.8; the text truncates. Table 2 prints no standard errors, so this figure appears only in prose
Reanalysis, p. 3	Whether the redefined-outcome interval excludes zero while the naive interval contains it	1	1.00000000	1	Evaluated here from the two p-values; in_text_claims.R evaluates the same contrast from the two confidence intervals. The deposit prints no p-values
Reanalysis, p. 3	Naive ATE, percentage points	-3.5	-3.54239000	1	-3.542 at full precision
Reanalysis, p. 3	Naive standard error, percentage points	2.0	2.04099000	1	2.041 at full precision
Reanalysis, p. 3	Estimates displayed in Table 2	4	4.00000000	1	Estimator rows in the reproduced Table 2

27 of the 31 checkable claims match the published article to the precision it prints, and both of the 2 descriptive claims hold. The 4 that do not match are the two bootstrap endpoints, which the R 3.6 sampler change explains and the old sampler restores, and the two figures in the reanalysis paragraph where the article's own prose disagrees with its own table and with the estimates behind it. Those two are the first two errata.

11 R environment

Item	Value
R version	4.6.2
Platform	aarch64-apple-darwin23
Date run	2026-11-12

Table 13: Package versions used for the run behind this report.

Package	Version
estimatr	2.0.1
rsample	1.3.2
dplyr	1.2.1
tidyr	1.3.2
purrr	1.2.2
readr	2.2.0
here	1.0.2